Quantification: 548_hyp_HP_z14c1-2
==================================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
----

measure	value
lacuna count	21
interior lacuna count	14
frame-edge lacuna count	7

Interior lacunae only: mean, median and sample SD (n = 14)
----

measure	kind	mean	median	sd	units
area_px2	lacuna shape	2580.29	1753	1930.27	px^2
major_axis_length_px	lacuna shape	78.7726	72.0264	32.9493	px
minor_axis_length_px	lacuna shape	43.3227	38.318	18.3368	px
aspect_ratio	lacuna shape	1.8566	1.9787	0.3968	unitless
eccentricity	lacuna shape	0.8092	0.8629	0.107	unitless
solidity	lacuna shape	0.7901	0.8075	0.0785	unitless
roots_count	headline	9.6429	7	5.9045	count
ring_length_r30_px	headline	311.357	274	155.238	px
ring_length_r60_px	reported	624.857	583.5	267.513	px
owned_length_px	ownership-dependent	1210.36	1140.53	617.606	px
edge_count	ownership-dependent	61.5714	56	33.9541	count
mean_edge_length_px	ownership-dependent	20.4328	19.6132	3.6505	px
perimeter_px	normalised	268.524	221.622	137.582	px
ring_area_r30_px2	normalised	8680.93	7623	3718.35	px^2
ring_area_r60_px2	normalised	19135.9	18514.5	7625.53	px^2
in_frame_fraction_r30	normalised	0.9773	1	0.0559	unitless
in_frame_fraction_r60	normalised	0.9533	1	0.0995	unitless
ring_density_r30	normalised	0.035559	0.03491	0.00597283	px^-1
ring_density_r60	normalised	0.0328808	0.0334308	0.00517482	px^-1
roots_per_100px_perimeter	normalised	3.7066	3.4779	1.3169	per 100 px
ring_attached_length_r30_px	network v2	289.071	238	152.086	px
ring_attached_length_r60_px	network v2	517	468.5	240.927	px
ring_length_w_r30_px	network v2	357.031	322.635	177.443	px
ring_length_w_r60_px	network v2	713.813	662.15	306.639	px
sholl_crossings_r10	network v2	7.4286	6.5	3.1553	count
sholl_crossings_r20	network v2	8.4286	8	4.4672	count
sholl_crossings_r30	network v2	8	6.5	4.4549	count
loop_count	loops, report only	4.2857	3.5	3.1968	count
loop_edge_share	loops, report only	0.2367	0.2199	0.096	share
ring_width_mean_r30_px	width	7.7283	7.7467	0.5235	px

Per field
----

measure	value	units
analysed_area_px2	991765	px^2
total_skeleton_length_px	28225	px
canalicular_length_density_per_px	0.0284594	px^-1
junction_count	615	count
junction_density_per_px2	0.00062011	px^-2
lacuna_count	21	count

lacunae_per_px2	2.117e-05	px^-2
skeleton_component_count	374	count
mean_component_length_px	75.4679	px
median_component_length_px	18	px
field_length_density_w_per_px	0.0324004	px^-1
field_density_without_flagged_per_px	0.0284594	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	180	px
unattached_component_count	11	count
canalicular_length_density_attached_per_px	0.0284594	px^-1
field_length_density_with_unattached_per_px	0.0286409	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	83	count
loop_edge_share	0.2168	share
edge_count	1573	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.855	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	28225	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2147	91.4001	39.1166	2.3366	0.9038	0.7373	0.5027	31.0005	542	True
2	2407	118.141	31.4268	3.7592	0.964	0.7314	0.0789	45.1836	688.173	True
3	3074	97.1472	47.5979	2.041	0.8717	0.7951	0.7084	40.3224	458.502	False
4	4688	114.763	56.7615	2.0218	0.8691	0.8409	-0.5667	73.3957	882.268	False
5	996	48.8421	30.1974	1.6174	0.786	0.83	-0.3922	177.571	118.507	False
6	1527	72.7312	32.1897	2.2595	0.8967	0.7689	1.1031	215.125	119.538	False
7	791	45.4603	27.379	1.6604	0.7983	0.777	-1.1935	276.899	15.5398	True
8	885	49.967	25.4783	1.9612	0.8602	0.8157	0.1758	321.858	27.7831	False
9	865	60.9988	23.7735	2.5658	0.9209	0.67	-0.6206	306.805	864.442	False
10	5590	100.499	74.9477	1.3409	0.6662	0.8931	0.3458	390.83	706.214	False
11	3806	88.412	63.5037	1.3922	0.6958	0.7996	0.5855	456.034	447.32	False
12	1368	68.8013	32.6693	2.106	0.8801	0.7621	0.1327	437.326	13.0395	True
13	818	48.7035	23.1908	2.1001	0.8794	0.8503	0.5594	490.792	650.099	False
14	583	30.9882	25.7454	1.2036	0.5565	0.8807	-0.6814	486.81	163.602	False
15	6173	146.612	73.4479	1.9961	0.8655	0.639	0.4437	581.134	750.702	False
16	1268	55.1715	37.4316	1.4739	0.7346	0.792	0.9741	664.745	233.587	False
17	1408	71.6194	28.9459	2.4743	0.9147	0.8167	1.0982	723.629	987.69	True
18	3872	116.659	53.0473	2.1992	0.8906	0.6702	-1.541	793.571	878.724	False
19	3821	95.6808	52.75	1.8139	0.8343	0.9065	-1.1603	884.495	983.684	True
20	1979	71.3216	39.2045	1.8192	0.8354	0.8154	-1.307	916.707	906.786	False
21	8745	138.777	89.9937	1.5421	0.7612	0.8115	1.2047	976.542	708.753	True

Network per lacuna

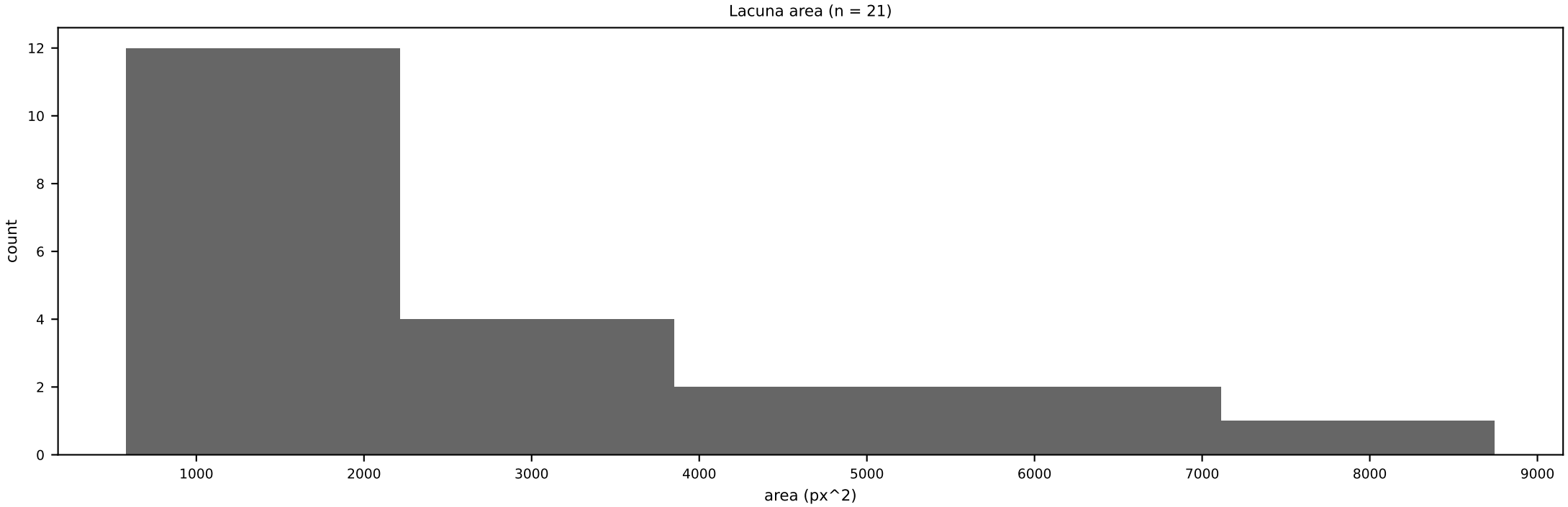
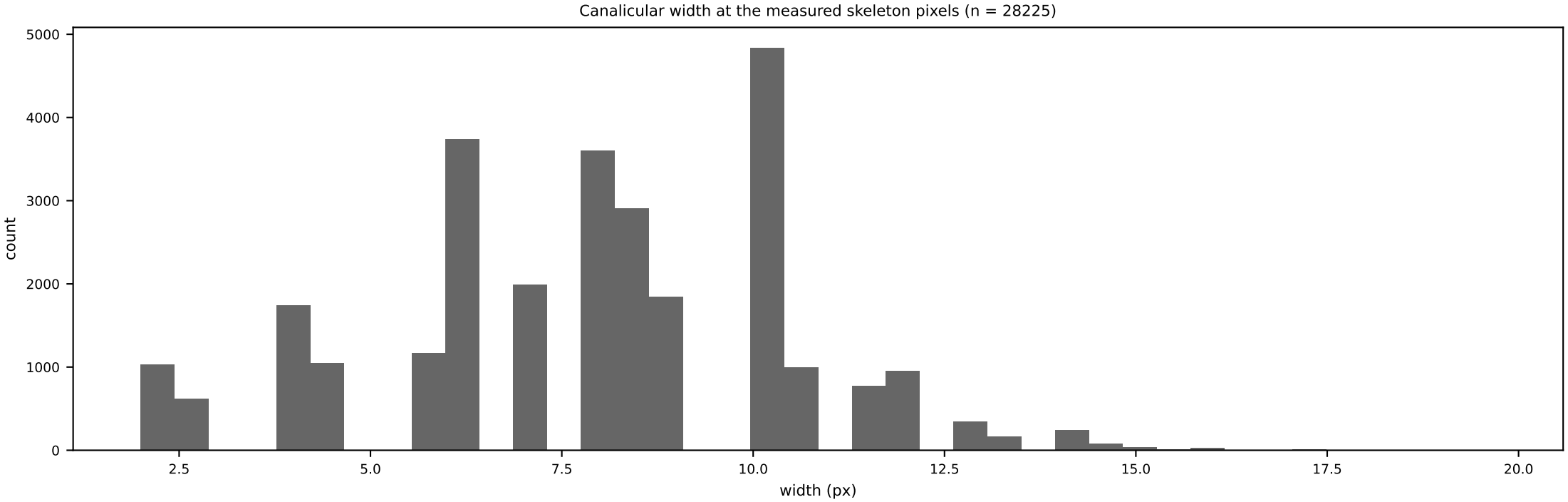
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	8	213	365	542.86	27	20.1059	6.6139
2	10	474	872	1523.14	73	20.8649	7.2057
3	10	252	520	1823.23	95	19.1919	7.8142
4	7	286	597	907.761	31	29.2826	8.5823
5	5	127	250	559.032	26	21.5012	7.8394
6	6	177	448	621.872	23	27.0379	7.3392
7	4	172	250	99.4975	5	19.8995	6.1619
8	6	162	302	411.333	21	19.5873	7.1062
9	11	323	691	1426.28	80	17.8285	7.1368
10	9	475	967	2100.6	109	19.2716	8.6069
11	12	395	885	1145.73	54	21.2173	7.712
12	2	123	257	184.368	5	36.8735	5.9051
13	4	275	451	1200.5	58	20.6983	7.4875
14	7	185	579	1870.46	102	18.3379	7.9722
15	26	661	1182	2293.38	116	19.7705	7.0262
16	7	273	588	1001.6	51	19.6391	7.404
17	8	311	574	1424.43	70	20.3489	7.755
18	18	541	877	1135.32	68	16.6959	7.7814
19	11	235	409	533.76	33	16.1745	7.0928
20	7	227	411	447.978	28	15.9992	8.3875
21	9	475	944	842.98	45	18.7329	6.6355

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	314.877	6967	12578	0.7247	0.6734	0.0305727	0.0290189	2.5407
2	330.693	8703	19577	0.7916	0.7301	0.054464	0.0445421	3.0239
3	289.563	8243	15900	0.836	0.748	0.0305714	0.0327044	3.4535
4	351.161	12196	26656	0.9903	0.8855	0.0234503	0.0223965	1.9934
5	192.989	4006	9474	1	1	0.0317025	0.026388	2.5908
6	221.38	6390	14742	1	1	0.0276995	0.0303894	2.7103
7	143.74	3169	5742	0.6661	0.6142	0.0542758	0.0435388	2.7828
8	148.61	4205	7189	0.8558	0.7127	0.0385256	0.0420086	4.0374
9	184.652	7467	20230	1	1	0.043257	0.0341572	5.9572
10	352.919	11920	25165	1	1	0.039849	0.0384264	2.5502
11	328.233	11389	28087	1	1	0.0346826	0.0315092	3.6559
12	193.539	5102	10952	0.6366	0.5949	0.0241082	0.023466	1.0334
13	137.196	6367	12630	1	1	0.0431915	0.0357086	2.9155
14	104.569	5596	16799	1	1	0.0330593	0.0344663	6.6942
15	631.595	16663	33826	1	1	0.0396687	0.0349435	4.1166
16	199.865	7779	21047	1	1	0.0350945	0.0279375	3.5024
17	197.48	6605	13178	0.8037	0.7245	0.0470855	0.0435574	4.051
18	394.747	12775	22894	1	1	0.0423483	0.038307	4.5599
19	281.38	6112	10015	0.7455	0.6918	0.038449	0.0408387	3.9093
20	221.865	6537	13263	1	1	0.0347254	0.0309885	3.1551
21	525.044	12017	25181	0.7555	0.698	0.0395273	0.0374886	1.7141

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	203	339	232.882	410.806	5	4	5
2	434	760	548.245	1009.55	7	16	11
3	210	435	288.693	589.174	6	7	4

4	276	490	334.191	695.039	9	9	10
5	127	186	143.397	273.823	4	4	2
6	157	337	200.095	508.132	4	5	6
7	142	182	199.681	287.522	2	3	4
8	162	266	188.995	352.363	6	3	4
9	293	582	380.99	799.838	7	10	6
10	475	932	544.76	1094.85	9	13	8
11	344	568	440.09	1012.38	11	8	17
12	73	108	129.426	280.966	2	3	8
13	211	349	327.363	533.671	5	10	4
14	167	447	207.924	661.014	4	4	10
15	634	958	760.625	1354.35	12	16	14
16	255	516	317.907	663.286	6	8	7
17	306	518	353.25	648.63	5	8	9
18	515	837	610.902	1001.34	14	17	14
19	215	363	268.693	463.475	7	6	7
20	221	335	252.51	454.12	7	4	6
21	423	675	532.032	1054.68	12	13	9

548_hyp_HP_z14c1-2: distributions, pre-validation, px